CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in data science with modern AI, from predictive models and time-series analysis to LLM features with RAG, prompt engineering and systematic evaluation of AI output quality. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by data science work on real process data at Covestro and a strong foundation in Python, pandas, scikit-learn and SQL.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Designed and optimized multi-agent LLM extraction (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, evaluating output quality against an SAP-system-derived ground truth; precision/recall raised from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Sole developer owning the full knowledge-extraction pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, letting domain experts validate and approve knowledge-graph changes.
- Built with Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Vector Search, MCP, Postgres, SQL, Neo4j, MongoDB, FastAPI, Docker, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, then designed and deployed an agentic LangGraph application that routes each request to vector database queries, general inference or summarization.
- Developed a React frontend and a Python backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, React, FastAPI, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.

- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Vector Databases, AWS (Lambda, SQS, DynamoDB, CDK)**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM predictive models for prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) for the tree-based models, while feeding raw sequences directly into the LSTM.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, pandas, scikit-learn, XGBoost, TensorFlow/Keras, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time-series analysis and forecasting of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, scikit-learn, TensorFlow**

PROJECTS

Project: Matching Experiment Lab | A/B Testing | statsmodels | XGBoost | RAG | Streamlit

- Job-matching experiment on synthetic healthcare-recruitment data: XGBoost ranking vs logistic baseline, LLM job-title normalisation with RAG and an output-quality eval, and experimentation on the new ranking: A/B tests with power analysis, sample size, significance testing and CUPED, reported as conversion funnels and KPIs in a Streamlit dashboard.

TECHNICAL SKILLSET

- **Data Science:** Python, pandas, NumPy, SciPy, scikit-learn, XGBoost, statistics and statistical modeling and time series (ARIMA, SARIMA, Prophet, LSTM), feature engineering, Matplotlib, Streamlit dashboards
- **LLM & RAG:** Retrieval-Augmented Generation, prompt engineering, evaluation of AI output (ground truth, RAGAS, PytrecEval), LangChain, LangGraph, LiteLLM, embeddings, vector databases
- **Data & Cloud:** SQL, Postgres, MongoDB, Neo4j, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure, Docker, Git, GitLab CI/CD, Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - LLMs Mastery : Transformers & Generative AI